

Berry curvature and response functions from VASP wavefunctions

VASPBERRY 1.3 scientific examples

25 September 2026

Abstract

VASPBERRY calculates Berry curvature, topological indices and response properties directly from VASP wavefunctions. Its main workflow reads a WAVECAR with the native Fortran executable and writes numerical results for post-processing and plotting, without constructing a Wannier model. The examples pair Fukui occupied-bundle Chern and Z_2 calculations with Kubo curvature on Brillouin-zone meshes and symmetry paths. Native interband matrix elements also support chemical-potential and temperature scans through Python occupation weighting and integration. Circular optical selection and real-space spinor densities complete the direct-wavefunction demonstrations.

Monolayer and bilayer MoS_2 , a Bi bilayer and a three-septuple-layer MnBi_2Te_4 film illustrate these capabilities and their numerical limits. VASP supplies the electronic structure; its band energies provide context for VASPBERRY's topology and response results. Fukui integer invariants and pointwise Kubo integrals answer different numerical questions: an integer invariant does not establish a converged transport integral. Optional physical-operator extensions and VASP-derived Wannier checks are documented separately in the appendices. The report documents software capabilities and numerical limitations with reproducible inputs, commands and reference results.

1. Methods

1.1 Berry curvature from wavefunction overlaps

The Fukui–Hatsugai–Suzuki method evaluates the Berry phase around each cell of a periodic $\mathbf{k}$ mesh using overlaps between neighboring wavefunctions. For an isolated group of bands, the overlap is a matrix and its determinant gives a gauge-invariant loop phase. With the connection $\mathbf{A} = i\langle u | \nabla_{\mathbf{k}} u \rangle$ used here,

$$\Phi_p = -\arg(U_1(\mathbf{k})U_2(\mathbf{k} + \Delta\mathbf{k}_1)U_1(\mathbf{k} + \Delta\mathbf{k}_2)^{-1}U_2(\mathbf{k})^{-1}), \quad \overline{\Omega}_p = \frac{\Phi_p}{\Delta S_k}.$$

The normalized link U_j is the phase of the overlap determinant. $\Delta S_k = |\mathbf{b}_1 \times \mathbf{b}_2|/(N_1N_2)$ is the area of a mesh cell, with reciprocal vectors containing 2π . Thus Φ_p is a phase and $\overline{\Omega}_p$ is an area-averaged curvature in $\text{\AA}^2$. The Chern number is $C = (2\pi)^{-1} \sum_p \Phi_p$. An isolated occupied group can include internal degeneracies; its separation from excluded bands remains essential. The discrete construction and its band-group extension are described by Fukui, Hatsugai and Suzuki.

The map and integral convey different information. A material can have strong local curvature at opposite valleys while its total Chern number is zero. The link construction supplies the lattice Chern invariant; the Kubo integral is not rounded or adjusted to reproduce that integer. An integer lattice Chern result also needs a sufficiently resolved mesh and a physically appropriate isolated band group.

1.2 Native Kubo curvature and Hall post-processing

For an isolated band and a Cartesian Hamiltonian derivative $D_\alpha = \partial H / \partial k_\alpha$,

$$\Omega_{n,xy}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

This form resolves curvature at each sampled $\mathbf{k}$ point. Sharp curvature near small gaps makes the $\mathbf{k}$ sampling and intermediate-band window particularly important. Individual-band values are not used where the sampled bands cannot be resolved separately. The native WAVECAR implementation uses canonical momentum of the stored pseudo-wavefunctions; full material velocity can require PAW, nonlocal and SOC terms. The exported-matrix interface permits an explicitly supplied physical operator (Appendix A).

The ordinary WAVECAR route is the starting point for the charge-Hall examples and requires no VASP source modification. Two optional comparisons extend the operator description: standard VASP optical output for insulating occupied bundles, and an instrumented producer for full velocity and spin matrices. These input routes and their distinct scopes are summarized in the operator guide. A more complete operator addresses an operator approximation; its $\mathbf{k}$ -mesh and band-window convergence must still be tested.

For an isolated occupied bundle $\mathcal{V}$, the trace curvature is

$$\Omega_{xy}^{\mathcal{V}} = -2 \text{Im} \sum_{n \in \mathcal{V}} \sum_{m \notin \mathcal{V}} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

Internal occupied-pair terms cancel in the sum of band curvatures. Excluding them before division avoids singular individual-band terms at internal degeneracies. The bundle must remain separated from the excluded bands. This occupied-to-unoccupied formulation and its numerical advantage are given by Wang et al., Eq. (11) and Sec. III D.

For a two-dimensional system, the intrinsic charge sheet conductivity is

$$\sigma_{xy}(\mu, T) = -\frac{e^2}{\hbar} \sum_n \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} f(E_{n\mathbf{k}} - \mu, T) \Omega_{n,xy}(\mathbf{k}).$$

Within an insulating gap at zero temperature, this becomes $\sigma_{xy} = -(e^2/h)C_{\text{occ}}$ in the stated convention. General background on Berry curvature and electronic transport is given in Xiao, Chang and Niu. The Bi Hall example evaluates the occupied-group flux directly; it does not assign independent curvatures to unresolved Kramers partners.

For chemical potentials crossing band edges, the Kubo implementation uses unordered interband pairs. With $N_{nm}^{xy} = -2 \text{Im}(D_{x,nm} D_{y,mn})$, the occupation-weighted integrand is $\sum_{n < m} (f_n - f_m) N_{nm}^{xy} / (E_n - E_m)^2$. Equal-occupation pairs cancel before division. Changes from a gap reference are evaluated with occupation differences before summation. This preserves small doping responses without subtracting large filled-band baselines. The temperature enters only the Fermi function; the electronic structure remains fixed. Regional contributions integrate the same charge response over specified parts of the BZ, with a $\mathbf{K}$ - $\mathbf{K}'$ difference defined without a factor of one-half. Such a partition is not a separately conserved valley-current operator.

1.3 From a WAVECAR to reusable results

The direct-wavefunction workflow has four steps:

1. Calculate the electronic states with VASP, using a full periodic mesh for BZ integrals or an explicit path for local curvature and band context.
2. Read the WAVECAR with the native Fortran executable. Choose Fukui links for lattice topology, Kubo matrix elements for pointwise curvature and charge response, or the optical and wavefunction output modes.
3. Retain the original numerical samples. For charge transport, the Python pair processor applies occupations, temperature and BZ or regional weights to native matrix elements; this is part of the response calculation. The common Python interfaces provide CSV, DAT and NPZ results where supported, with units and represented bands recorded.
4. Plot those outputs beside the material's VASP band energies. Display interpolation may smooth a map, but does not replace the sampled data used for integration or convergence checks.

The current repository build provides the following `--task` selectors:

Native task	Numerical output
chern	Fukui occupied-bundle flux, curvature and Chern number
z2	Fukui–Hatsugai integer field and half-zone parity
kubo	Pointwise band or bundle curvature on a mesh or path
kubo-pairs	Interband-pair data for Python charge-Hall integration
optical / spectrum	Circular transition selectivity / broadened spectra
wavefunction	Selected Γ -state amplitudes for density plots

For example, `build/vaspberry --task kubo` selects the native pointwise calculation; supply the WAVECAR and band options listed in the usage guide and worked examples. The task aliases refer to the current repository version; the immutable v1.3.0 release uses the retained legacy flags.

These stages require no Wannier localization. Python prepares inputs, normalizes outputs and performs the stated post-processing. Optional standard-WAVEDER, supplied physical-matrix and Wannier interfaces use additional Python numerical backends with input requirements described in Appendix A and Appendix B.

2. Materials and numerical settings

Dataset	Wavefunctions and sampling	Quantities shown
Monolayer MoS ₂ , full zone	SOC; 12×12 mesh; 26 bands; 400 eV	Fukui and Kubo occupied-bundle curvature, bands 1–18
Monolayer MoS ₂ , matching path	SOC; 49 K–Γ–K′ points; 26 bands; 400 eV	VASP band structure and Kubo curvature beside the maps
Monolayer MoS ₂ , local valleys	SOC; two 9×9 patches; 26 bands; 400 eV	Isolated band-18 curvature near K and K′
Monolayer MoS ₂ , transport	SOC; 12×12 to 36×36 meshes; stored and retained band windows varied separately; 400 eV	Chemical-potential and temperature dependence of regional Hall response
Monolayer MoS ₂ , operator comparison	Same 12×12 SOC eigenstates; 60 stored bands; pairs within 1–40 or 1–50; 400 eV	Ordinary WAVECAR and optional full PAW velocity Hall curves
MoS ₂ stacking comparison	Monolayer and 1H, 2H, 3R bilayers; PBE+SOC; separate 6×6 SCF densities; 49 Γ–M–K–Γ–K′ points; 60/64 path bands; 400 eV	VASP bands and circular optical selection from WAVECAR and standard WAVEDER
Monolayer MoS ₂ , supplied path	SOC; 48 K–Γ–K′ points; 32 bands; 400 eV	Γ-point state density; original optical tutorial
Buckled Bi bilayer	Two atoms; PBE+SOC; 400 eV; fresh 12×12 SCF density; full 6×6, 12×12 and 18×18 meshes; occupied bands 1–10	Native Z ₂ n-field; optional PAW spin response in Appendix A
MnBi ₂ Te ₄ , three septuple layers	SOC+U; 21 atoms; full 6×6 VASP mesh; 192 bands; occupied bands 1–123; 270 eV	Occupied WAVECAR Fukui invariant; coarse optical-integral diagnostic

The input guide links the structures and VASP files. The MoS₂ maps and their matching path use the same public SCF charge density, structure, potentials, cutoff and 26-band window. VASP 5.4.4 generated the 12×12 mesh and 49-point path with ICHARG=11 and ISYM=-1; only the k-point list changes. The sampled occupied-to-empty direct gap is 1.674 eV. The tutorial supplies preparation files and commands for both WAVECARs. The stacking examples use fixed, documented idealized structures and a common 24 Å cell height. The 3R slab additionally uses a self-consistent z dipole correction. These small path calculations illustrate optical analysis across input systems; no structural or full-zone response convergence is asserted. The stacking guide provides the geometries and complete VASP workflow. The real-space example and original optical tutorial retain the separate supplied 32-band path.

The optional Bi and MnBi₂Te₄ Wannier representations and their separate interpolation checks are described in Appendix B. Their model construction is additional to the WAVECAR workflows in the main text.

3. Results from VASP wavefunctions

3.1 MoS₂: Fukui Berry curvature over the full Brillouin zone

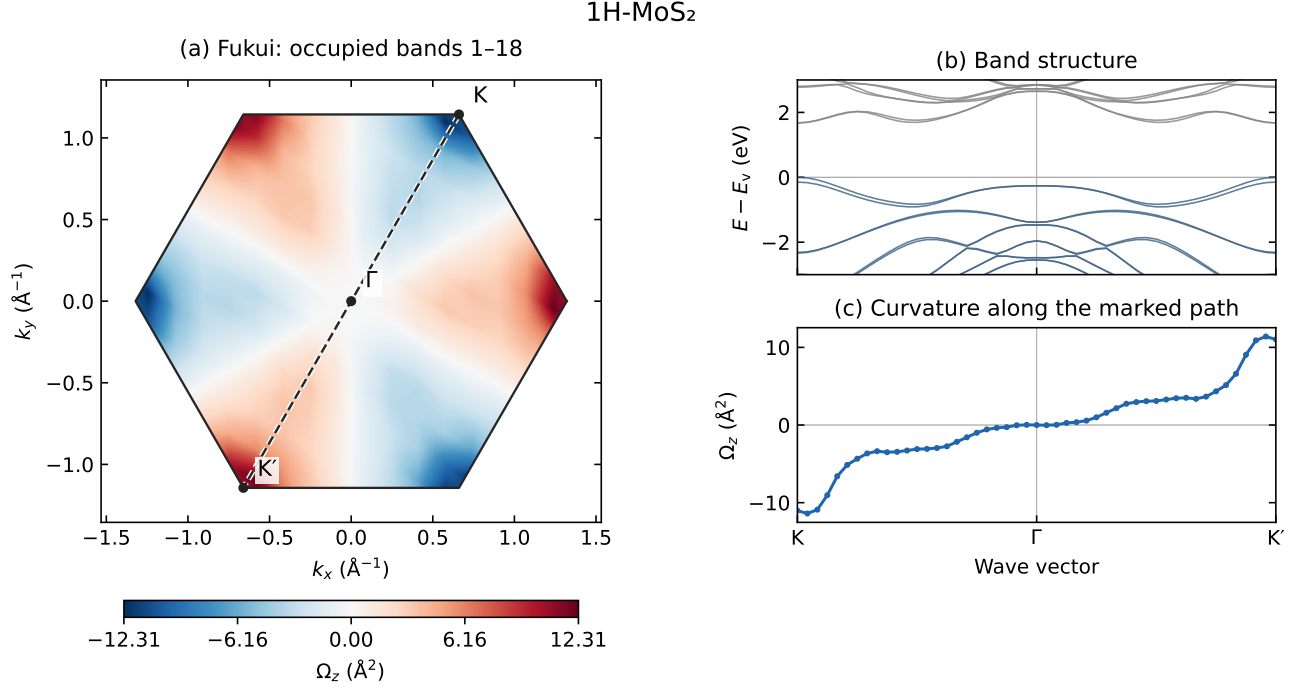


Figure 1. (a) Fukui Berry curvature of occupied bands 1–18 in the Cartesian first Brillouin zone. The dashed line marks K– Γ –K', with $K = (1/3, 2/3)$ and $K' = -K$ in reciprocal coordinates. Native 12×12 plaquette values are displayed with periodic bilinear interpolation onto a 401×401 Cartesian grid. (b) VASP band structure along that path; energies are relative to the valence-band maximum, with occupied bands in blue and empty bands in gray. (c) A periodic bilinear cut of the plaquette field along the marked path. This line visualizes panel (a), rather than adding an independent pointwise curvature calculation. Both right-hand panels share the same Cartesian path distance.

The curvature has opposite signs near the time-reversed K and K' valleys. The largest sampled magnitudes are approximately **12.31** $\text{\AA}^2$, while the full-zone Chern number is zero to numerical precision. The native text map integrates to approximately -5×10^{-7} after decimal rounding. This illustrates the distinction between valley-contrasting local geometry and vanishing total charge Chern number. The connection between inversion breaking, spin-orbit coupling and valley optical response in MoS₂ is discussed by Xiao et al.. The present map is a finite-mesh example rather than a convergence study.

Calculation and plotting commands

3.2 MoS₂: Kubo curvature of the occupied valence-band bundle

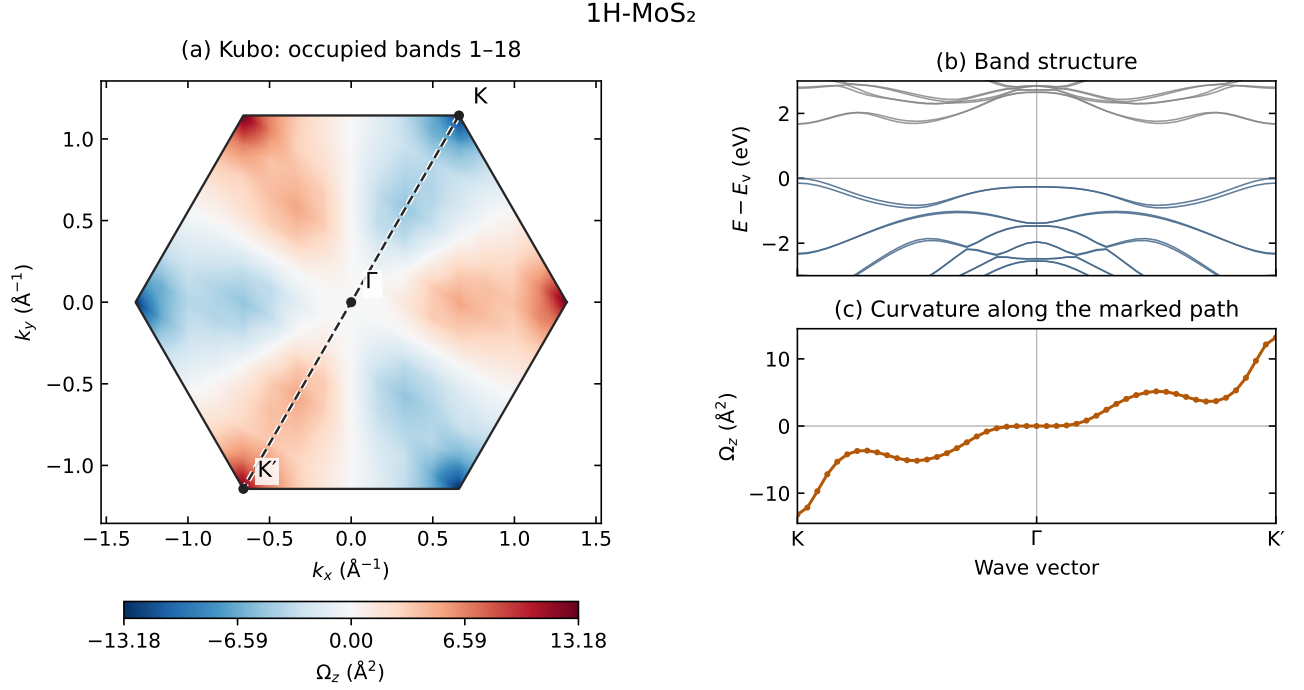


Figure 2. (a) Canonical-momentum Kubo trace curvature of occupied bands 1–18 on the 12×12 mesh. Only transitions to the stored empty bands 19–26 enter the sum. The map uses the same periodic bilinear display interpolation and dashed K – Γ – K' path as Figure 1. (b) VASP band structure, with occupied bands in blue and empty bands in gray. (c) Bundle curvature calculated directly at all 49 path points. Internal valence-band degeneracies do not interrupt the bundle curve.

The occupied-bundle curvature is approximately $-13.18 \text{ \AA}^2$ at K and $+13.18 \text{ \AA}^2$ at K' . The occupied space stays separated from the empty states by at least **1.674 eV** on both samplings, so every mesh and path point is valid for the bundle calculation. The native option `-kubo_bundle 1` performs the external-band sum directly, avoiding large cancelling internal terms.

Figures 1 and 2 describe the same occupied band space. Their numerical values need not coincide at this resolution: Fukui gives finite-plaquette averages, while this Kubo result uses pointwise canonical momentum and a finite empty-band window. The 12×12 mesh and 26 stored bands require convergence for quantitative predictions. NumPy interpolation smooths only the display; all integration and numerical checks use the original samples.

Calculation and interpretation

3.2.1 MoS₂: an isolated single band near the valleys

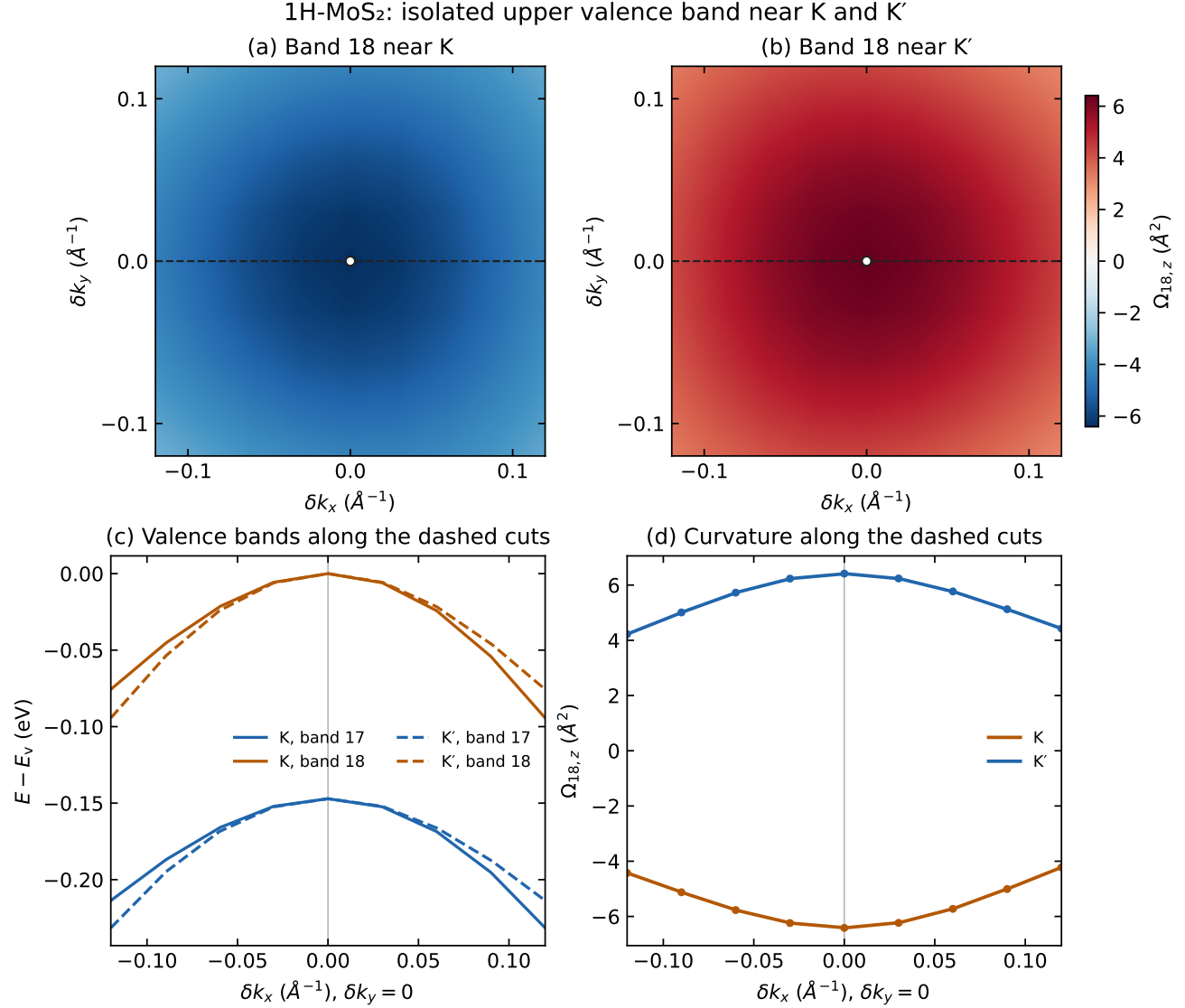


Figure 3. Single-band Kubo curvature of the upper valence band, band 18, around K and K'. Each local Cartesian patch extends $\pm 0.12 \text{ \AA}^{-1}$ and contains 9×9 VASP points. The upper maps use NumPy bilinear display interpolation; the dashed lines mark the cuts below. The lower panels show the two upper VASP valence bands near both valleys and band-18 curvature at both valleys. Symbols mark the calculated curvature samples.

Band 18 touches its Kramers partner at Γ , but is separated from every other stored band by at least **0.130 eV** throughout these valley patches. The 17–18 splitting at K is **0.147 eV**, and the valley curvatures are approximately **$-6.415/+6.415 \text{ \AA}^2$ at K/K'**. This is a suitable local single-band example of the spin-split valley physics described by Xiao et al.. The patch data describe local curvature; a full-zone integral requires the appropriate isolated band bundle and complete BZ sampling.

VASP preparation, calculation and reference data

3.2.2 MoS₂: intrinsic charge and regional valley Hall response

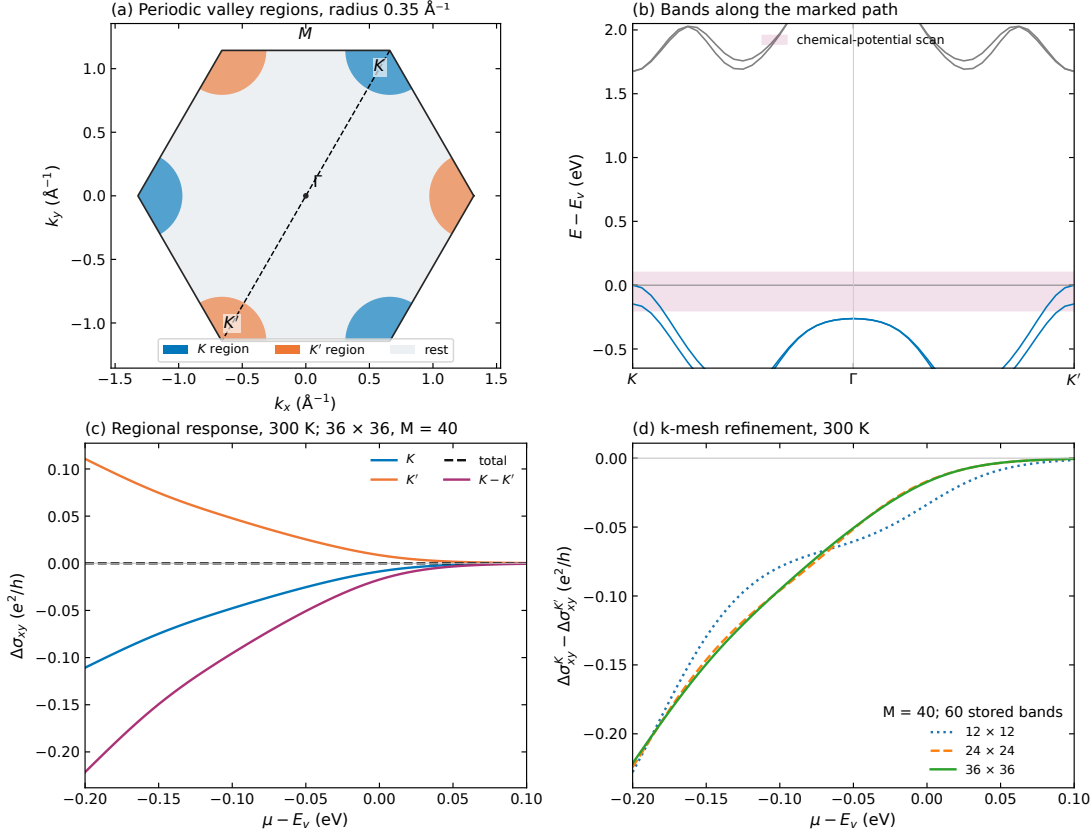


Figure 4. (a) Cartesian first BZ with periodic K/K' disks of radius $0.35 \text{ \AA}^{-1}$ and the marked K- Γ -K' line. (b) The matching 26-band VASP dispersion, with the chemical-potential interval shaded. (c) Regional Hall changes at 300 K on the 36×36 mesh, using 60 stored bands and pairs within bands 1–40. (d) Mesh refinement at that fixed band window. Energies are relative to each source's valence-band maximum; the band path uses the same density and cutoff.

The native Fortran calculation supplies the interband pairs; Python post-processing applies occupations and integrates their contribution over the full zone and the specified regions. The 61-point scan covers $E_v - 0.20$ to $E_v + 0.10$ eV at 0 and 300 K. We define $\Delta\sigma(\mu) = \sigma(\mu) - \sigma(\mu_{\text{ref}})$ with a midgap reference, and the valley difference as $\Delta\sigma_K - \Delta\sigma_{K'}$. The total charge response remains below $2.24 \times 10^{-9} e^2/h$, consistent with time-reversal symmetry. Numerical tables also give absolute conductivities and carrier counts.

Two numerical choices were varied separately at 300 K. The relative L2 difference below is the norm of the curve change divided by the finer curve's norm, on a common $\mu - E_v$ grid.

Test	Refinement	Relative L2 change
k mesh; 60 stored bands, pair cutoff 40	$12 \times 12 \rightarrow 24 \times 24$	12.89%
k mesh; same band window	$24 \times 24 \rightarrow 36 \times 36$	1.482%
Pair cutoff; one 12×12 , 96-band source	$60 \rightarrow 80$	1.263%
Pair cutoff; same source	$80 \rightarrow 90$	0.202%

The final mesh change exceeds the stated 1% criterion. The last cutoff increment is below 1%; these separate tests do not establish joint convergence. The reference guide includes the commands and additional checks on empty-state accuracy and numerical degeneracy grouping.

This rigid-band response uses canonical momentum, omitting PAW augmentation and nonlocal/SOC velocity terms. Numerical refinement does not remove those approximations. The supplementary plot retains the finite-mesh steps at 0 K.

3.2.3 Optional comparison: matching the current operators

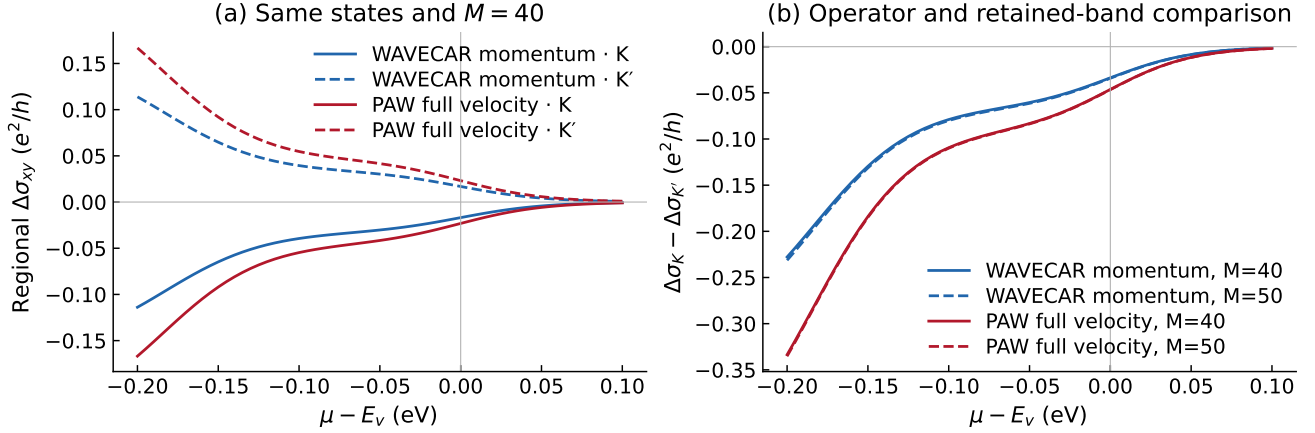


Figure 5. Charge response from the same final VASP eigenstates on a 12×12 mesh, with 60 stored bands, at 300 K. (a) Regional changes for K and K' using pairs within bands 1–40. (b) The K–K' difference with retained cutoffs $M = 40$ and 50. Blue curves use ordinary WAVECAR canonical momentum; red curves use the optional full PAW velocity matrices. Geometry, density, occupations, region definitions and eigenstates are identical.

At $\mu - E_v = -0.20$ eV and $M = 40$, the regional difference changes from -0.22779 to -0.33363 e^2/h when the full velocity replaces canonical momentum. Both total charge responses remain below 1.1×10^{-8} e^2/h , without time-reversal averaging. The nonzero regional response therefore reveals an operator difference that the symmetry-enforced total cannot test.

For the complete 300 K difference curve, increasing M from 40 to 50 changes the canonical and PAW results by **1.445%** and **0.465%**, respectively, using the same relative-L2 definition as above. At $M = 50$, the two operators differ by 29.44% relative to the PAW curve. These measurements describe one fixed mesh and band increment. They do not establish k convergence or a general advantage in convergence rate.

The full-velocity route includes the supported PAW, nonlocal and SOC operator terms, giving a more complete current description. Generating its input uses the optional VASP instrumentation; ordinary WAVECAR remains the starting workflow. The separate comparison guide provides preparation instructions, both reusable pair caches and the original CSV/DAT/NPZ results. Their integration can be repeated without VASP.

3.3 MoS₂: circular optical selection across layer stackings

The native optical calculation resolves left- and right-circular transitions from the WAVECAR momentum matrix elements. Circular selectivity is their intensity difference divided by their sum. The same analysis can be repeated with standard VASP optical matrices as an optional comparison.

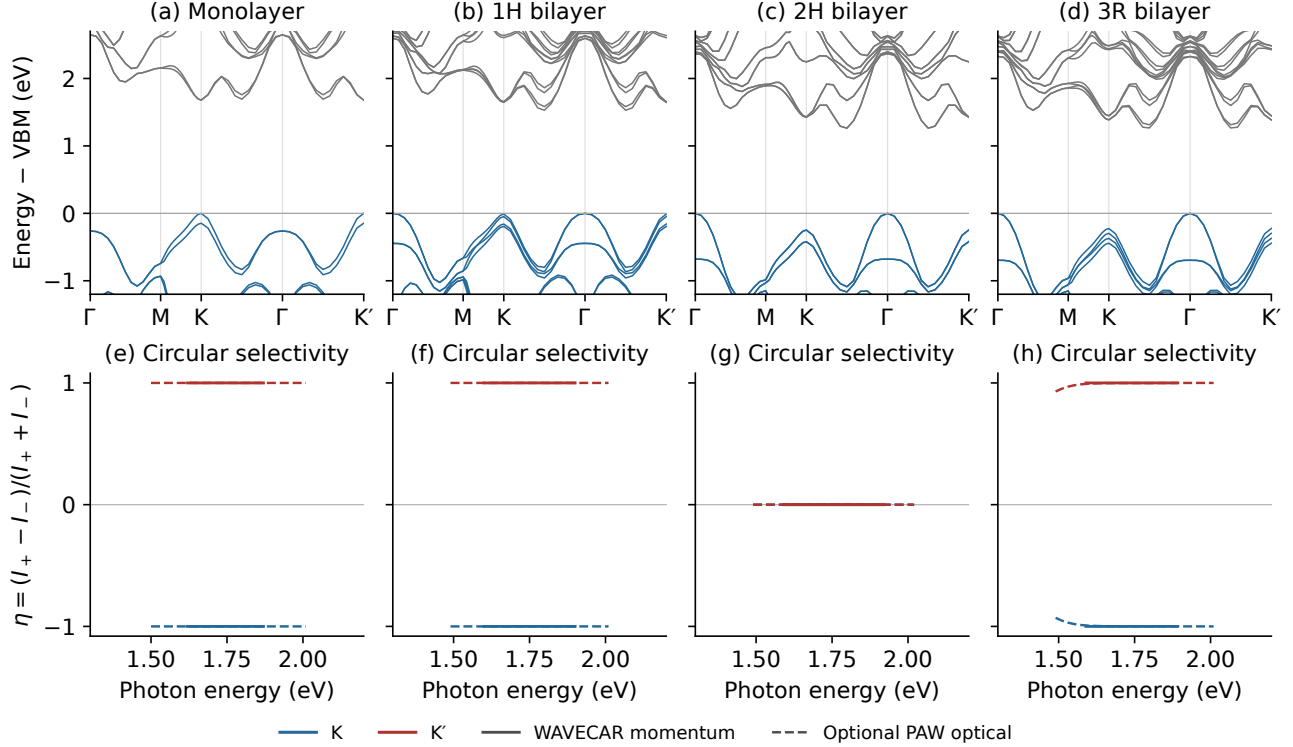


Figure 6. (a–d) VASP bands of monolayer and 1H, 2H and 3R bilayer MoS₂ along Γ –M–K– Γ –K', with each case referenced to its own sampled VBM. (e–h) K/K' circular selectivity under normal incidence, from ordinary WAVECAR momentum (solid) and optional standard WAVEDER matrices (dashed). Transitions use occupied bands 1–18 to 19–20 for monolayer, and 1–36 to 37–40 for all bilayers, with Gaussian $\sigma = 0.05$ eV. Weak spectra and native ratios limited by printed intensity precision are masked; overlapping curves retain their calculated values.

The near-edge peaks lie at **1.67–1.68 eV**. Monolayer and 1H show $\eta \simeq -1/+1$ at K/K'; the 3R PAW peak gives **–0.99884/+0.99884**. For the inversion-symmetric 2H control, summing the complete occupied and empty groups cancels the selectivity: the PAW peak residual is below 5×10^{-7} at both valleys. This illustrates why the response of an unresolved degenerate group must include every partner.

The stacking comparison extends one optical workflow to several VASP input systems. It is motivated by the 1H bilayers reported by Yang et al., using fixed idealized geometries for the present software demonstration. It does not reproduce the paper's direct-gap prediction or measured photoluminescence polarization. Earlier figures already demonstrate curvature and Hall integration; those maps are not repeated for each stacking.

The optional PAW optical calculation uses ordinary VASP output and requires no source modification. Its comparison with native momentum tests helicity and symmetry. The two spectral definitions have different finite-broadening weights, so their ratio differences do not isolate an operator correction and their absolute intensities are not equated. The stacking guide and reference spectra provide complete VASP inputs, both channel spectra and reproduction commands; the original monolayer optical tutorial remains the introductory example.

3.4 Bi: native Fukui–Hatsugai Z_2 topology

The two-atom buckled Bi bilayer provides a compact real-material example of quantum spin Hall topology, as proposed by Murakami. We use the fixed Bi geometry and a fresh nonmagnetic PBE+SOC calculation, with a 400 eV cutoff and electronic tolerance of 10^{-8} eV. The native calculation reads the resulting VASP spinor wavefunctions directly from WAVECAR.

VASPBERRY evaluates the Fukui–Hatsugai lattice n-field from the occupied SOC bands 1–10 on a full Γ -centered 12×12 mesh. With the time-reversal-compatible gauge used in this construction, the Z_2 invariant is the parity of the integer-field sum over a half Brillouin zone:

$$\nu = \left[\sum_{p \in B_{1/2}} n(p) \right] \bmod 2.$$

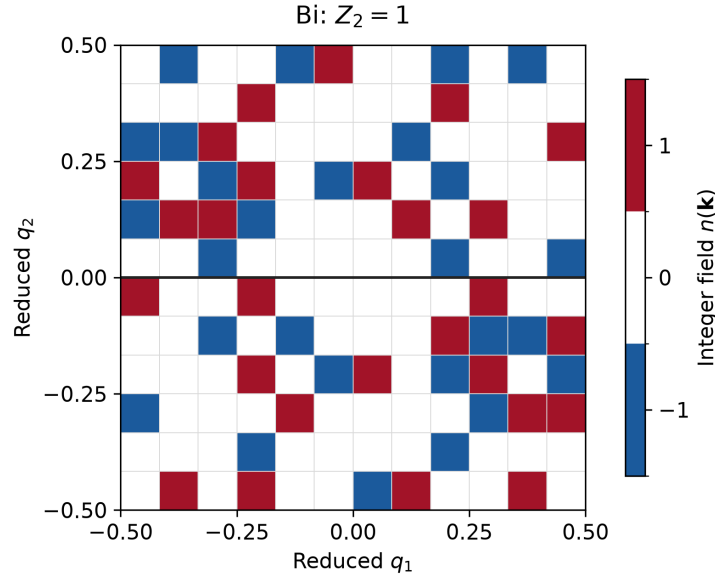


Figure 7. Fukui–Hatsugai integer field $n(\mathbf{k})$ calculated from the Bi VASP spinor wavefunctions. Red, white and blue denote +1, 0 and -1, respectively. The native plaquettes are displayed without interpolation in dimensionless reduced coordinates, $\mathbf{k} = q_1 \mathbf{b}_1 + q_2 \mathbf{b}_2$, with opposite edges periodically identified. The line at $q_2 = 0$ separates the upper and lower half zones. The occupied-band calculation gives $Z_2 = 1$.

The upper and lower half-zone sums are **-3 and +3**, respectively. Both are odd, so their parities agree at $\nu = 1$, identifying the nontrivial time-reversal-symmetric insulating phase for this calculation. The sampled minimum direct and global gaps are **0.535 eV** and **0.497 eV**.

The local n-field depends on the gauge and logarithm branch; its half-zone parity is the invariant. The reduced-coordinate map shows the original mesh half-zones used for these sums. The separate occupied $\mathbf{C} = \mathbf{0}$ calculation and zero charge-Hall response are consistent with this nontrivial Z_2 result.

The native topology calculation uses WAVECAR pseudo-wavefunction overlaps. An independent check of the original occupied subspaces, before the native TR reconstruction, gives a maximum time-reversal residual of 4.57×10^{-8} . The optional spin Hall calculation in Appendix A.3 uses physical spin and velocity matrices with the PAW terms described in Appendix A.2. The older Bi fixture remains a separate historical reference.

Actual Bi preparation and commands · Numerical n-field

The native occupied-subspace calculation establishes the bulk Z_2 result shown here. Optional edge connectivity corroborates it in Appendix B.2; conventional spin Hall conductivity is a separate observable and need not be quantized when spin is not conserved (Appendix A.3).

3.5 MoS₂: a real-space Γ state

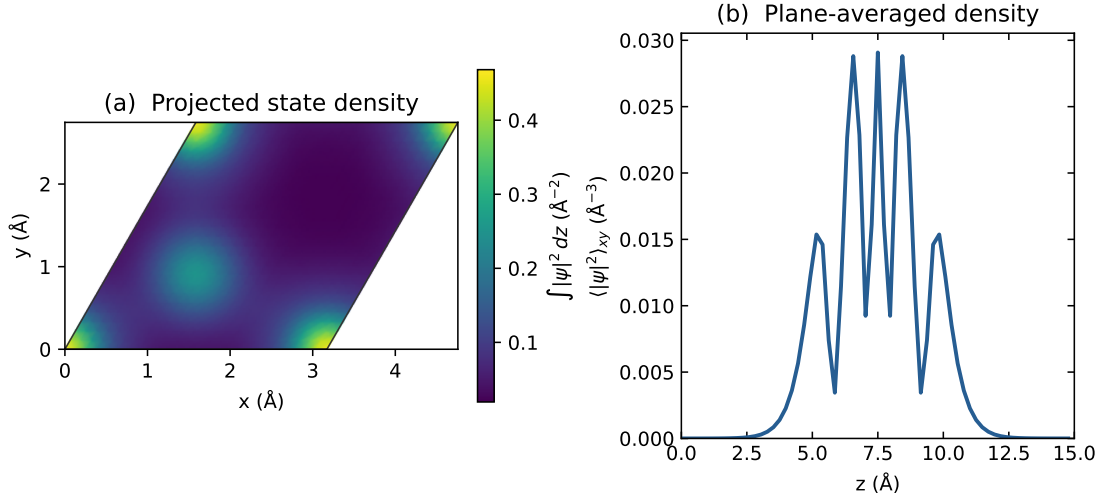


Figure 8. Projected pseudo-wavefunction density of band 18 at Γ and its plane average along z . The in-plane plot uses the Cartesian geometry of the oblique primitive cell. Both spinor components are retained.

The integrated pseudo-density is approximately **0.8258**, consistent with the plane-wave coefficient norm. The PAW augmentation is not part of this plot, so a unit density integral is not imposed. The example illustrates conversion of complex amplitudes into a physical-coordinate density map.

Wavefunction calculation guide

3.6 MnBi₂Te₄: occupied-bundle Fukui Chern number

This example demonstrates a nonzero Chern invariant of the complete occupied VASP band space. The three-septuple-layer MnBi₂Te₄ film has 21 atoms with +z/-z/+z Mn magnetization. Its fixed, unrelaxed geometry retains bulk vertical spacings from Yan et al., with in-plane $a = 4.336 \text{ \AA}$, PBE+SOC and Mn $U_{\text{eff}} = 5.34 \text{ eV}$ following Otrokov et al. It is distinct from the relaxed films in that study.

The VASP gap is **17.10 meV**. Fukui links of the **complete occupied valence-band bundle, bands 1–123, give $C = -1$** , corresponding to $\sigma_{xy} = +e^2/h$ in our convention. A direct 6×6 PAW optical integral instead gives **163.109 e^2/h** : the Γ -point curvature reaches approximately $-1.52 \times 10^4 \text{ \AA}^2$, and this coarse mesh overweights the narrow peak. This optical result is a convergence diagnostic, not a quantized response. An integer Fukui result does not establish convergence of the pointwise Kubo integral, and no integer value is imposed on that integral.

The native Fortran calculation evaluates overlap determinants directly in the complete VASP occupied space. Its $C = -1$ result and all 36 plaquette fluxes agree with the independent Python wavefunction-Fukui reference within the native map’s printed precision. The separately checked deep bands 1–36 form a $C = 0$ bundle. The native output and comparison and material guide provide the actual command, VASP inputs, overlap diagnostics and standard-WAVEDER coarse-mesh result.

The main lesson is the distinction between occupied-bundle topology and response integration. The optional dense Wannier calculation in Appendix B.3 checks the expected Hall value within a supplied finite model; it is not needed to perform the direct-wavefunction workflows above and does not establish convergence of this coarse VASP optical integral.

4. Practical use

Begin with the example guide, reproduce the chosen quantity, and then select the sampling and band window appropriate to your own material. Use Cartesian reciprocal-space maps to display lattice symmetry, named k paths for band-resolved quantities, and energy or chemical potential for spectra and transport. State units, represented bands, temperature and broadening in the caption.

Start with WAVECAR and the native executable for Fukui topology, Kubo curvature, optical selection and state densities. For charge Kubo transport, export the native interband pairs and use the Python occupation/integration step to scan chemical potential and temperature without repeating the matrix-element calculation. These main workflows do not require Wannier90. Optional standard optical or full transition-matrix inputs allow additional operator terms to be assessed on matching electronic states. The operator guide links the input requirements and separate instructions for each route.

For quantitative conclusions, converge the VASP electronic structure, k mesh and relevant band window. The examples show how the methods are used; the supplementary analytic references check formulas and conventions, while validation details document the numerical implementation. Detailed execution records remain alongside the numerical files for reproducibility.

Appendix A. Optional physical-operator extensions

These additional routes address the operator content of response functions. They complement the native WAVECAR examples; they are not prerequisites for Fukui topology or canonical-momentum Kubo analysis. Standard WAVEDER needs no VASP source change. Full velocity and PAW spin inputs use the separately documented instrumented producer. Their Python backends and convergence conditions are explicit below; the matched charge comparison remains in Section 3.2.3 so readers can assess its effect on the native baseline.

A.1 Standard VASP optical matrices

For an insulating occupied bundle, the supported standard VASP longitudinal optical calculation supplies matrix elements $C_{cv,\alpha} = \langle u_c | \partial_{k_\alpha} u_v \rangle$ in Å. Here v and c label occupied and empty states. Their contribution to the trace is

$$\Omega_{xy}^{\text{occ}} = -2 \text{Im} \sum_{v,c} C_{cv,x}^* C_{cv,y}.$$

The energy denominator is already contained in these derivatives. The longitudinal PAW optical expression includes projector and augmentation terms, as described by Gajdoš et al.. This standard WAVEDER route needs no VASP source modification. The `waveder-hall` command checks the actual VASP output, occupied filling and global gap before integration. Its present scope is VASP 5.4.4, zero temperature and a fixed insulating bundle; the occupied–empty separation must exceed the producer’s 2 meV degeneracy threshold. Mesh and empty-state convergence remain separate requirements.

The same standard optical matrices provide circular transition spectra at each $\mathbf{k}$ point. For propagation along $+z$ and $\epsilon_\pm = (\hat{x} \pm i\hat{y})/\sqrt{2}$,

$$I_\pm(\mathbf{k}, E_\gamma) = \sum_{v,c} \left| \frac{C_{cv,x} \pm iC_{cv,y}}{\sqrt{2}} \right|^2 g_\sigma(E_\gamma - E_c + E_v), \quad \eta = \frac{I_+ - I_-}{I_+ + I_-}.$$

Here g_σ is a normalized Gaussian. Complete initial and final band groups are summed before forming η , which is essential at degeneracies. These $\mathbf{k}$ -resolved strengths are independent-particle quantities; they do not include excitons or emission kinetics. The ordinary WAVECAR calculation uses canonical momentum with its native photon-energy normalization. The optical guide specifies both conventions and their comparison.

A.2 PAW spin matrices and conventional spin Hall response

The conventional spin current is $J_i^a = \{s_a, v_i\}/2$, with $s_a = \hbar\sigma_a/2$. In SOC systems its matrix elements require the full complex spin matrix, including off-diagonal band elements. Multiplying charge Berry curvature by a spin expectation value is generally insufficient. This current definition and its first-principles evaluation are discussed by Qiao et al. and Ryoo, Park and Souza.

The two WAVECAR spinor components directly give the pseudo matrix $\tilde{\Sigma}_{nm}^a = \sum_{Gss'} c_{nGs}^* (\sigma_a)_{ss'} c_{mGs'}$. The physical PAW matrix additionally requires the on-site correction

$$\Sigma_{nm}^a = \tilde{\Sigma}_{nm}^a + \sum_{Aijss'} p_{ni,s}^{A*} \Delta Q_{ij}^A (\sigma_a)_{ss'} p_{mj,s'}^A, \quad \Delta Q_{ij}^A = \langle \phi_i^A | \phi_j^A \rangle - \langle \tilde{\phi}_i^A | \tilde{\phi}_j^A \rangle.$$

Here p denotes the same-run PAW projector overlap. This expression is $T^\dagger \sigma_a T$ for the supported spin-independent PAW transformation. The implementation checks the corresponding physical overlap and preserves the raw pseudo-state norms. For this optional route, the supplied producer recipe instruments a separately licensed VASP 5.4.4 source copy. The ordinary WAVECAR and standard WAVEDER charge workflows do not need that instrumentation.

For transport, $D_i = \hbar v_i$ includes PAW and nonlocal/SOC velocity terms. The producer retains the velocity before the optical energy-denominator division, including diagonal and exactly degenerate blocks. These blocks cannot be recovered from an optical connection that has already discarded them. Within the supplied source-band space P , VASPBERRY forms $K_i^a = \{P\Sigma^a P, PD_i P\}/2$. This finite-band product omits the $P\Sigma^a QD_i P$ and $PD_i Q\Sigma^a P$ terms, where $Q = 1 - P$. Increasing the source-band window therefore tests both current-product closure and the intermediate-state sum.

For the complete occupied group $\mathcal{V}$, the implementation evaluates

$$\Omega_{ij}^a = -2 \operatorname{Im} \sum_{n \in \mathcal{V}, m \notin \mathcal{V}} \frac{K_{i,nm}^a D_{j,mn}}{(E_m - E_n)^2}, \quad \frac{\sigma_{ij}^a}{(\hbar/e)(e^2/h)} = \frac{A_{BZ}}{4\pi} \sum_{\mathbf{k}} w_{\mathbf{k}} \Omega_{ij}^a(\mathbf{k}).$$

Internal equal-occupation terms cancel before division. The formula uses $e > 0$, physical electron charge $-e$, Pauli matrices without an embedded $\hbar/2$, and D in eV Å. The curvature-like tensor is in Å². In a conserved $\sigma_z = +1$ sector, this normalized spin coefficient is minus one half of the charge coefficient in Section 1.2. No additional spin degeneracy or integer rounding is applied. The present command supports a fixed insulating 2D bundle at zero temperature. Its operator, gauge, units, sampled gap and band coverage are recorded with the numerical output.

Z_2 topology and conventional spin Hall response are separate tests. A nontrivial time-reversal invariant insulator need not have an exactly quantized bulk spin Hall coefficient when spin is not conserved. The optional ideal-edge check in Appendix B.2 addresses boundary connectivity, not a finite-device conductance. The spin Hall guide gives the operator contract and reproduction commands.

A.3 Bi: optional spin Hall conductivity and convergence

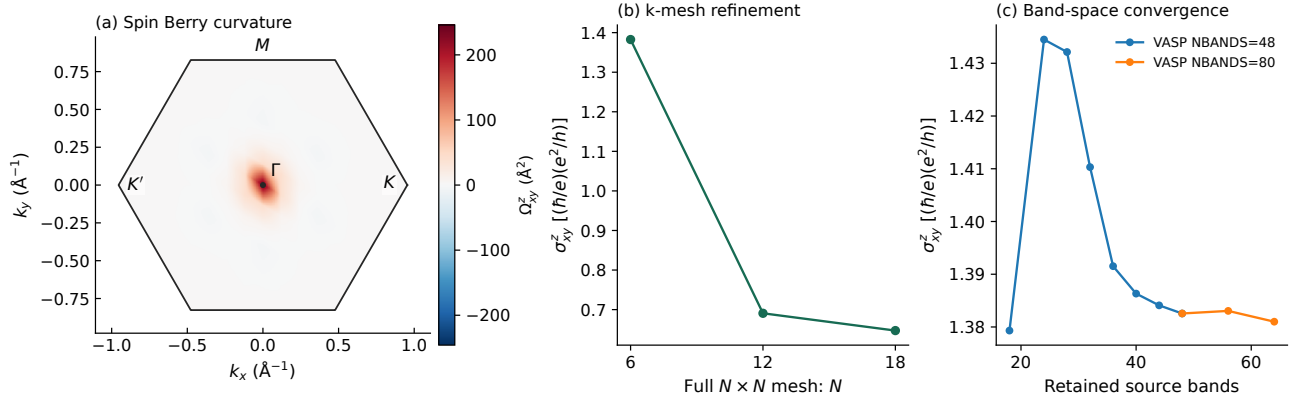


Figure 9. (a) Occupied-bundle spin Berry curvature Ω_{xy}^z from PAW spin and full velocity matrices on the 12×12 VASP mesh, shown in the Cartesian first BZ. Periodic bilinear interpolation is used only for the color map. (b) Mesh refinement retaining 48 bands from 64-band VASP sources. (c) Retained-band studies from 48- and 80-band VASP sources at a fixed 6×6 mesh. The numerical integrals always use the original k points and weights.

Retaining 48 states from 64-band VASP calculations, the 6×6 , 12×12 and 18×18 meshes give $\sigma_{xy}^z = 1.382558$, 0.690723 and 0.647020 in $(\hbar/e)(e^2/h)$. The last change is 0.04370 , or 6.75% of the 18×18 value. **These meshes do not establish a converged spin Hall conductivity.** The main difficulty is the narrow response near Γ : its local value stays near $245.9 \text{ \AA}^2$, but its weighted contribution falls from 1.286413 to 0.321603 and 0.142934 . The nearest sampled radius falls from 0.276 to 0.138 and $0.092 \text{ \AA}^{-1}$. Smoothing the plot cannot supply the missing integration resolution.

The band study probes a different error. At fixed 6×6 sampling, retaining 40, 44 and 48 bands from the 48-band source gives 1.386328 , 1.384098 and 1.382555 . Increasing the actual VASP source to 64 bands while retaining the same first 48 changes the response by only 2.64×10^{-6} . The uppermost new empty states fail a Kramers-pair accuracy check. An 80-band VASP run resolves the first 64 states: retaining 48, 56 and 64 gives 1.382558 , 1.383047 and 1.381001 , with a final change of 0.15% . Only the verified subspace is used; an electronic stopping criterion alone does not establish accuracy of every stored empty state. Neither band stability nor an integer Z_2 invariant establishes k convergence.

The 12×12 charge response is $1.82 \times 10^{-7} e^2/h$, consistent with the vanishing anomalous charge Hall effect of this nonmagnetic system. With 64 source bands and the accurate first 48 retained, that residual falls to $-2.29 \times 10^{-12} e^2/h$; the 18×18 residual is $4.91 \times 10^{-9} e^2/h$, without time-reversal averaging. The finite spin response uses a different current vertex and is allowed by time reversal. Its noninteger value is not a failed Chern-number calculation. The finite-band current-product approximation, conventional-current definition, sampling and operator scope remain explicit in the output.

The supplied actual 6×6 and 12×12 PAW matrix bundles reproduce the VASPBERRY integration without rerunning VASP. The fresh SCF density, VASP input preparation and original producer insertion routines document the earlier stages; the additional Wannier inputs belong to the separate check in Appendix B.2. Every response component was independently recomputed from the raw exported matrices; all 27 spin components agree to within 3.1×10^{-14} in the stated sheet units. A separate finite- k check of the diagonal velocity gives a maximum error of $1.74 \times 10^{-4} \text{ eV \AA}$. The latter is a producer validation on development test points, not a material convergence result.

Spin Hall reproduction guide · Numerical conventions · Validation details

Appendix B. Optional Wannier supporting validation

The calculations in this appendix require an additional VASP-derived Wannier model. External Wannier90 prepares the representation; VASPBERRY imports the supplied Hamiltonian and position matrices and evaluates the stated quantities itself. It neither generates the VASP electronic states nor performs Wannier localization. These checks support bulk topology and response verification while the main user workflow remains direct WAVECAR analysis.

B.1 Full-connection evaluation of the supplied model

As a separate supporting route, VASPBERRY interpolates the Hamiltonian and position matrices of a VASP-derived Wannier model. Its own Fourier, diagonalization and occupied-trace routines evaluate $\Omega = J0 + J1 + J2$: the basis-connection curl, the mixed connection/Hamiltonian-derivative term, and the derivative-pair term. Only occupied-to-empty denominators occur; internal degeneracies are allowed. The formulation follows Wang et al. and Lopez et al., Eq. (51). Bands and local curvature are checked against direct VASP data. The separate postw90 calculation in Wannier90 provides an independent check of VASPBERRY's evaluation of the same finite model.

B.2 Bi: ideal-edge support for the native bulk Z_2 result

The auxiliary model uses 16 s/p spinor orbitals and a 6×6 VASP training mesh; independent 12×12 and 18×18 VASP comparisons test its dispersion. Finite strips of 20 and 40 cells test the width dependence.

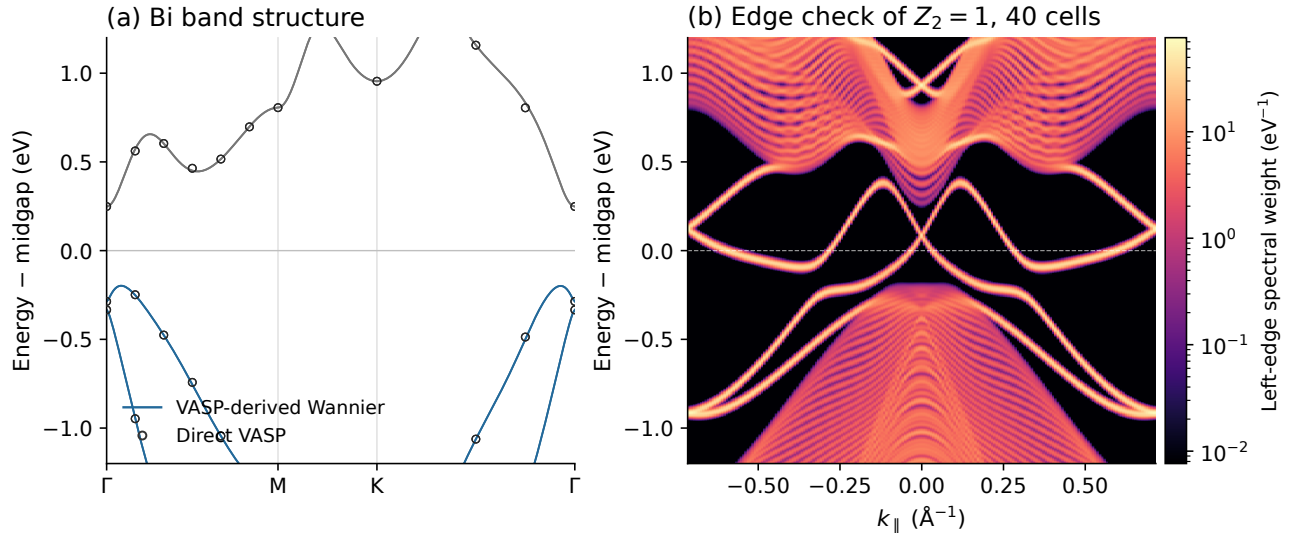


Figure 10. (a) Bi band structure from a VASP-derived 16-orbital Wannier Hamiltonian along Γ -M-K- Γ . Circles show direct VASP energies on the independent 12×12 mesh; all ten occupied bands are retained in the model. (b) Spectral weight in the first two cells of a 40-cell strip, periodic along the first lattice vector and open along the second. All eigenstates are summed before applying a 10 meV Gaussian display broadening. Energy zero is the midpoint of the sampled VASP bulk gap. The edge branches connect the valence and conduction manifolds across that gap.

The separation of the central Kramers doublets at Γ falls from 7.354 meV for a 20-cell strip to 0.03046 meV for 40 cells. Three crossings per edge occur over the positive half of the one-dimensional BZ at each of three test energies inside the bulk gap. This odd, edge-localized connectivity corroborates $Z_2 = 1$ and the time-reversal-protected boundary modes expected for this phase. The edge calculation is a supporting check of the bulk topology in Figure 7.

The VASP-derived interpolation was checked on separate 12×12 and 18×18 meshes: its sampled gaps differ from VASP by at most 2.426 meV and bands 9–12 by at most 33.23 meV. Localization stopped before its requested spread tolerance. The strip represents an ideal truncation of bulk hoppings, with no edge relaxation or device contacts. The Bi reproduction guide retains the complete interpolation, symmetry and strip-width checks.

B.3 MnBi₂Te₄: dense integration of a supplied finite model

Section 3.6 gives the complete occupied-wavefunction invariant $C = -1$ and the unresolved coarse optical integral. This optional calculation tests the corresponding Hall expectation with a VASP-derived Wannier representation.

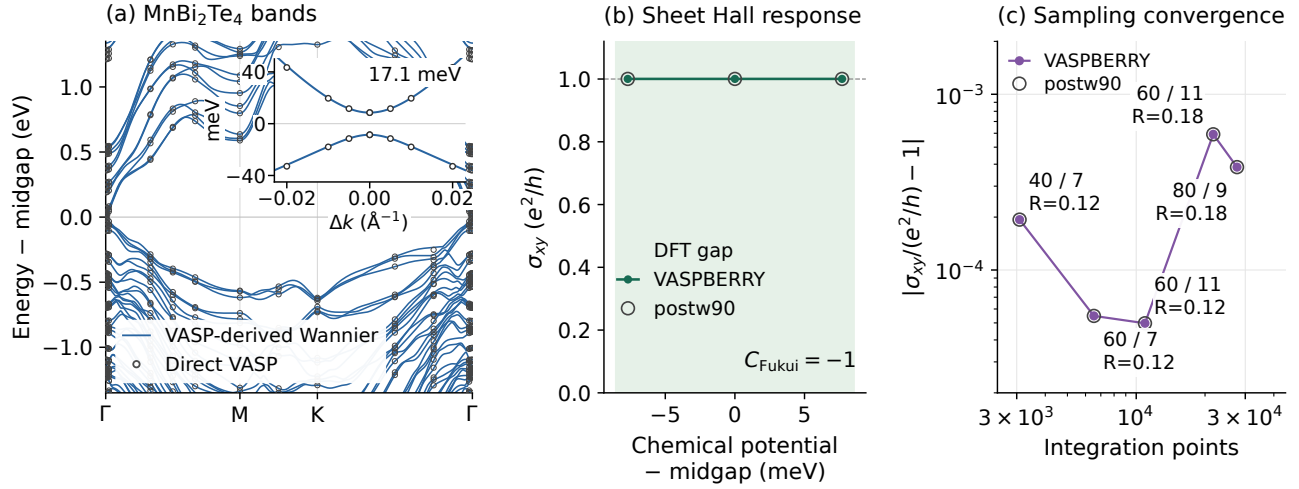


Figure 11. (a) VASP-derived Wannier band structure along Γ -M-K- Γ , with direct VASP checks; the inset resolves the K- Γ -M gap near Γ . (b) VASPBERRY full-connection sheet Hall response at three chemical potentials inside the gap, at zero temperature; open markers show the independent postw90 comparison. The dashed line is the topological expectation. (c) Deviation $|\sigma_{xy}/(e^2/h) - 1|$ under integration refinement. Labels give the base mesh, local subdivision and Γ -region radius in $\text{\AA}^{-1}$. Both solvers use the same model and weights; their agreement is distinct from mesh convergence.

For dense integration, VASPBERRY's `wannier-hall` evaluates all J0/J1/J2 terms from the supplied VASP-derived Hamiltonian and position matrices. `wannier-bands` evaluates the plotted interpolation of the same VASP-derived electronic structure. The model contains 138 orbitals and **87 occupied states**; the omitted 36 deep VASP bands form a separately checked $C = 0$ bundle. Near Γ , direct VASP band edges agree within 1.26 meV and local curvature within 1.44%; the latter comparison retains the full-123 versus model-87 subspace difference. The maximum band-edge difference across all twelve validation locations is 18.14 meV.

VASPBERRY's final 27,600-point result is $\sigma_{xy} = 1.000384978 e^2/h$ at all three gap chemical potentials. It uses an 80×80 base mesh and 9×9 subdivisions in the $0.18 \text{ \AA}^{-1}$ Γ region. The last refinement changes the result by $2.07 \times 10^{-4} e^2/h$, below the stated $10^{-3} e^2/h$ comparison criterion. The nonmonotonic region-expansion control is retained; no integer rounding is applied. Independent postw90 totals and all three component decompositions agree within their original output precision. A separate 160×160 energy scan places all three chemical potentials inside the sampled model gap.

This validates VASPBERRY's full-connection evaluation of a fixed finite model. The source 6×6 mesh, structure and basis still require convergence for quantitative material predictions; localization stopped before its spread tolerance. The material tutorial provides the actual inputs, VASPBERRY commands and outputs. The independent external reference is retained separately. This input route requires both Wannier operators; it does not infer missing PAW/nonlocal/SOC velocity terms from a WAVECAR alone.

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